Global, regional, and national health-care inefficiency and associated factors in 201 countries, 1995–2022: a stochastic frontier meta-analysis for the Global Burden of Disease Study 2023

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Summary

Background All governments face pressure to maximise the impact of their health budget. We aimed to measure health spending inefficiency for 201 countries from 1995 to 2022, estimate the cost of one additional year of healthy life, and assess contextual factors associated with health spending inefficiency.

Methods We extracted data from the Global Burden of Diseases, Injuries, and Risk Factors Study 2023 and the Financing Global Health 2024 project to estimate health spending inefficiency using a non-linear stochastic frontier meta-analysis model designed to assess health-adjusted life expectancy (HALE). This model produced a frontier that represents the best possible HALE for a given level of health spending. Inefficiency scores were measured as the distance between a country's HALE and the frontier at that country's level of spending. We used the slope of the frontier to estimate the cost of one additional year of healthy life, and we regressed inefficiency scores on contextual factors and policy variables to measure their association with health spending inefficiency.

Findings The relationship between health spending and HALE was positive for all levels of spending, although health spending inefficiency existed in most countries. Globally, health spending inefficiency decreased from 1995 to 2019, increased considerably in 2020 and 2021 due to the COVID-19 pandemic, and recovered substantially in 2022. We found decreasing returns to additional health spending, with the cost of one additional health-adjusted life-year varying from US$92 (95% uncertainty interval 43–239) per capita for a country spending $100 per capita to $11 213 (8031–57 754) per capita for a country spending $5000 per capita. More efficient spending was associated with better governance, having a higher percentage of health expenditure from the government, infrastructure that facilitates access to and delivery of health care, and higher uptake of preventive care measures.

Interpretation Expanding government-provided health-care coverage would decrease the inefficiency of the health-care system. Countries should also focus on strengthening democracy, building infrastructure, and increasing the use of, and access to, preventive care.

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Introduction

Health-care systems, especially in low-income countries, are facing unprecedented financial strain in the post-pandemic era. Post-pandemic debt coupled with higher interest rates has led to interest payments almost doubling for low-income countries since 2019.[1] Government budgets face additional pressure from competing priorities, including the increasing impacts of climate change. For countries that are reliant on development assistance, there is the additional pressure of recent cuts by the USA and the UK to development assistance for health.[2,3] The combination of these factors has led to an increasing focus on decreasing health spending inefficiency.

To quantify and assess health spending inefficiency, several recent studies have done data envelopment analyses (DEAs)[4] to compare efficiencies across countries and found substantial health-related and economic losses due to inefficiency.[5–9] A study by Eze and colleagues[9] included the highest number of countries to date, estimating health-care system efficiency for 191 countries in 2021. They focused on universal health care and found that countries could increase their coverage by about 5% with current resource levels. Goyeneche and Bauhoff[5] focused on life expectancy and found that countries in Latin America and the Caribbean could increase life expectancy by an average of 3·5 years by eliminating spending inefficiency.

There is also a growing literature that seeks to explain the variation in inefficiency by regressing inefficiency scores against possible drivers. Higher income inequality[6] and worse governance quality (eg, higher levels of corruption) have consistently been found to be associated

Research in context

Evidence before this study

We searched PubMed with the terms "health (spending OR systems) AND (stochastic frontier analysis OR data envelopment analysis)" for studies published from database inception to April 15, 2025. Several recent studies have done a data envelopment analysis to evaluate the inefficiency of health spending across countries using either life expectancy or universal health coverage as outputs, highlighting potential gains through decreased spending inefficiency. These studies have also examined factors associated with spending inefficiency, including governance quality, income inequality, and health-care system characteristics such as hospital beds per capita and vaccination rates. To the best of our knowledge, no studies have examined health spending inefficiency up to 2022 or estimated the cost of additional health outputs.

Added value of this study

This study models health spending inefficiency for 201 countries using total health expenditure as the input to the health-care system. A frontier for health-adjusted life expectancy is estimated with a novel non-linear methodology—a stochastic frontier meta-analysis. We estimated inefficiency over 28 years and produced the first inefficiency estimates that extend to 2022. We found that the cost of one additional health-adjusted life-year increases with health expenditure, indicating that additional spending will provide the largest gains for countries with small health-care budgets. Factors associated with more efficient spending were better governance, having a higher percentage of health expenditure from the government, infrastructure that facilitates access to health care (eg, access to electricity), and higher uptake of preventive care measures, such as antenatal visits.

Implications of all the available evidence

These findings underscore the potential for countries to achieve better health outcomes by focusing on decreasing health spending inefficiency. The study finds that well-functioning democracies, government expenditure on social protection, infrastructure that enables access to care, widespread use of preventive care, and consolidation of development assistance are associated with reduced inefficiency. For policy makers, these insights offer actionable strategies to optimise health outcomes in an era of post-pandemic recovery and tightening health budgets.

with less efficient health spending.[5,6] Factors found to be associated with more efficient health spending include a higher number of hospital beds per capita, more years of primary schooling, and higher childhood immunisation rates.[10]

See Online for appendix

In this work, we aimed to add to the previous literature on health spending inefficiency by estimating inefficiency for more country-years than previously done, spanning 201 countries and 28 years, from 1995 to 2022. This research also provides the first inefficiency estimates that include the COVID-19 era. We used a newly developed approach called stochastic frontier meta-analysis (SFMA).[11] This method improves upon both DEA and stochastic frontier analysis (SFA), which are commonly used methods of estimating a production frontier. We also examined novel indicators as possible drivers of inefficiency, including development assistance for health and spending on social protection. This study is also the first, to the best of our knowledge, to provide estimates of the cost of achieving an additional year of health-adjusted life expectancy (HALE) at all levels of health spending based on the slope of the frontier.

Methods

Overview

We extracted data from the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2023 and the Financing Global Health 2024 project to estimate an efficiency frontier for HALE. We used the slope of the efficiency frontier to estimate the cost of one additional health-adjusted life-year at all levels of health spending. We then analysed several policy and structural factors that we hypothesised would explain some of the variation in health spending inefficiency. A graphical representation of the steps of this analysis is provided in the appendix (p 35).

Stochastic frontier meta-analysis

Typical methods for estimating the efficiency frontier include SFA and DEA. DEA is a non-parametric approach for estimating inefficiency in which a piecewise linear frontier is estimated on the basis of the decision-making units with the best outcomes. Key limitations of both SFA and DEA are their inability to incorporate observation-level standard errors and to remove outliers. DEA has also been known to produce infeasible weight values on inputs and outputs.[12] A limitation of classic SFA is the need to impose assumptions on the shape of the frontier.

To address these limitations, Zheng and colleagues[11] developed the SFMA method. SFMA offers a non-parametric approach that captures non-linear frontiers with optional constraints on the shape. The outcome envelope is modelled with basis splines, and all coefficients are estimated in an all-at-once likelihood-based approach.

A key advantage of the SFMA approach is the ability to distinguish noisier from less noisy datapoints via user-specified data uncertainty. In the standard SFA approach, two error terms are estimated: a symmetric error term and a half-normal inefficiency term. In the SFMA approach, the model is specified with an additional error term, as follows:

$$y_i = \langle X_i, \beta \rangle + u_i - v_i + \epsilon_i$$
$$\epsilon_i \sim N(0, \sigma_i^2)$$
$$u_i \sim N(0, \gamma)$$
$$v_i \sim HN(0, \eta)$$

where β defines a spline on X_i, ϵ_i is the sampling error with known variance σ_i^2, u_i are random effects with unknown variance γ, and v_i are the inefficiencies with unknown variance η (appendix p 37).

SFMA is also robust to outliers, leveraging an extension of the least trimmed squares method to identify outliers in the input data. This method uses a computational approach to find a subset of points that minimises the sum of squared residuals.

Efficiency frontier model

The efficiency frontier represents the maximum HALE achievable on the basis of a given level of health spending. The efficiency score is the distance from each country's actual HALE and the efficiency frontier at that country's level of total health expenditure.

HALE for 1995 to 2022 was extracted from the GBD dataset.[13] Total health expenditure for 1995 to 2022, which includes public, private, and out-of-pocket expenditures, was extracted from the Financing Global Health Study.[14] Total health expenditure was measured in purchasing power parity-adjusted 2023 international dollars, adjusted to reflect country-specific purchasing power. To ensure comparability between countries, we controlled for two factors that affect health outcomes but are primarily influenced by factors distal to the health-care system. First, we adjusted for Socio-demographic Index (SDI).[15] The SDI is a composite measure of total fertility in women younger than 25 years, income, and education that reflects a country's socioeconomic development.[15] We also controlled for a measure of the contribution of a comprehensive set of environmental, behavioural, and metabolic risks to mortality known as the joint population attributable fraction (PAF).[16] More details on the joint PAF are provided in the appendix (p 34). Natural disasters and epidemics were not controlled for, even though they can have large impacts on health outcomes, due to their transient nature.

Countries that spent less than US$35 (2022) per capita on health care were excluded from the model as their spending levels and outcomes were systematically different from other countries; 1·8% of country-years were thus excluded from the model. The excluded countries and years are shown in the appendix (p 36). After this exclusion, 5% of the input data were trimmed during model fitting and the remaining 95% were used to construct the final frontier. The frontier model was estimated with 250 draws of total health expenditure, HALE, and all covariates, to incorporate uncertainty.

The 95% uncertainty interval (UI) was constructed from the 2·5th and 97·5th percentile draws of the frontier. The cost of achieving one additional health-adjusted life-year was calculated as the inverse of the slope of the frontier, and the units are total health expenditure per capita per health-adjusted life-year. This cost assumes that as a country increases its total health expenditure, its level of inefficiency remains the same. If a country were to increase spending while simultaneously reducing inefficiency, the cost per one health-adjusted life-year would decrease.

Association with health-care systems characteristics

We regressed inefficiency scores on several potential associated drivers of inefficiency, including factors related to health financing, governance, preventive care, and infrastructure variables. The categories included for this analysis were based on an informal review of the literature. Within each category, we aimed to include as broad a set of metrics as possible based on data availability.

Financing indicators included the proportions of health spending on out-of-pocket payments, government funding, prepaid private sources, and development assistance.[1] We also examined the fragmentation of development assistance. Fragmentation was measured as the Herfindahl–Hirschman Index, which measures market concentration with the market share of firms or

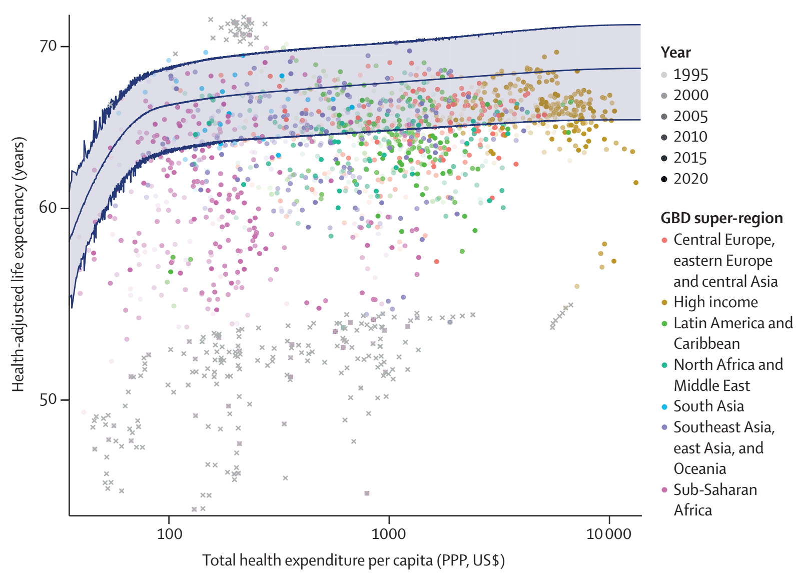

Figure 1: **Health-adjusted life expectancy, health spending per capita, and the efficiency frontier, 1995–2022**
The frontier is estimated by controlling for Socio-demographic Index, and an indicator of the fraction of disability-adjusted life-years of a country attributed to risk factors that were generally not determined by the health-care system, such as environmental and occupational risks, using 250 draws of each variable. Datapoints that are treated as outliers during the stochastic frontier meta-analysis modelling process are marked with an X. Uncertainly intervals, depicted by the shaded region, are based on the 2·5th and 97·5th percentile of these 250 estimates. The solid lines represent the upper bound, mean estimate, and lower bound. For readability, only data from 1995, 2000, 2005, 2010, 2015, 2019, and 2022 are shown. The year legend indicates that later years are more opaque, but the legend does not list all years shown. PPP=purchasing power parity.

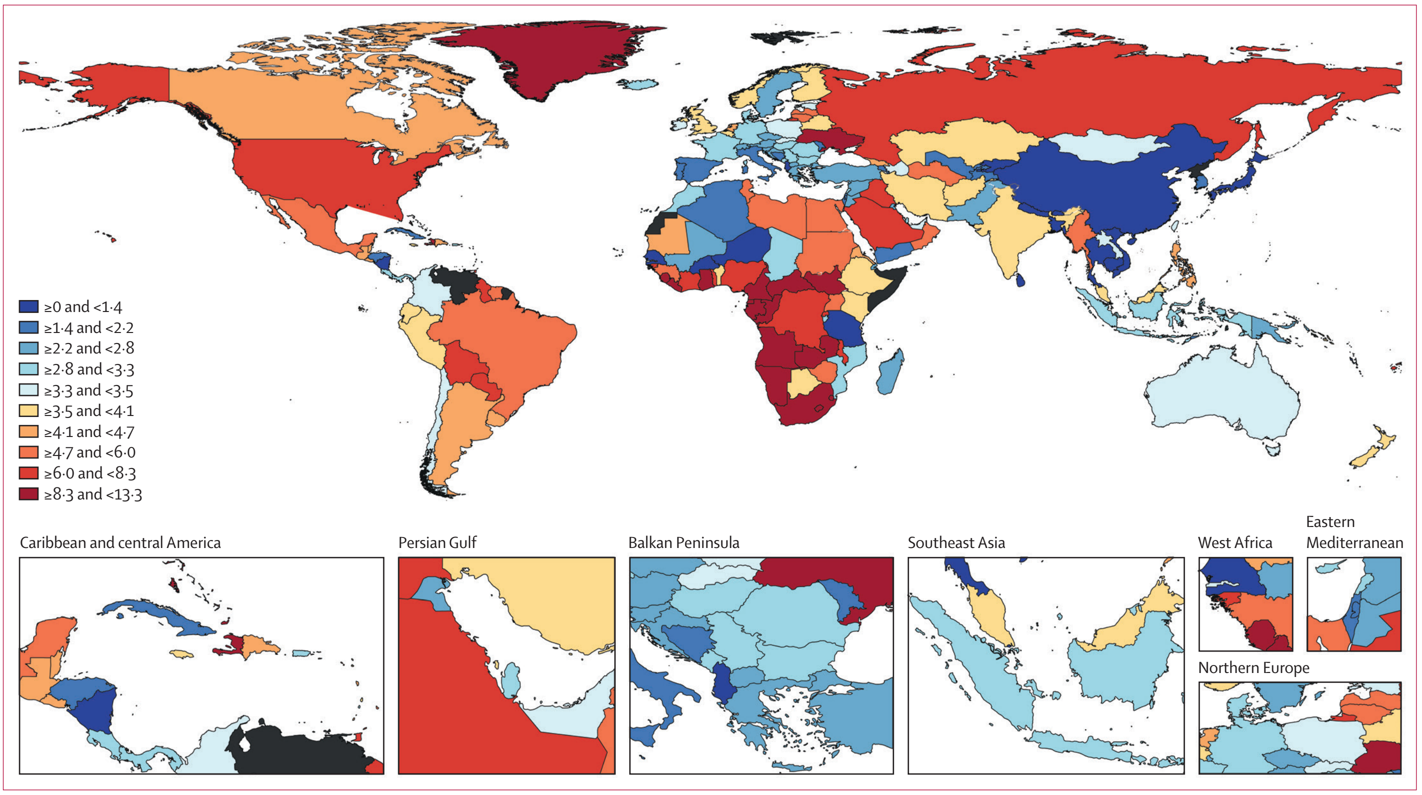

Figure 2: **Inefficiency scores, 2022**
The inefficiency score is the distance from each country's health-adjusted life expectancy (HALE) and the maximum HALE achievable on the basis of that country's level of total health expenditure (efficiency frontier). Locations without inefficiency estimates are shown in dark grey.

entities. In this study, the Herfindahl–Hirschman Index is created by treating each combination of one donor and one programme area as one entity having some market share of total development assistance. For example, Global Fund–malaria and Global Fund–tuberculosis would be two donor–programme areas, and they are treated as separate entities because funds are not transferrable between these two areas. Spending allocation was explored with data from National Health Accounts to create metrics for the percentage of spending on inpatient, outpatient, and pharmaceutical care.[17]

Governance indicators included the Worldwide Governance Indicators compiled by the World Bank, as well as the degree of democracy, the generosity of social protection, and the level of education inequality. A description of each metric can be found in the appendix (pp 8–12). Preventive care measures explored were vaccination rates, antenatal care, and skilled birth attendance. Infrastructure indicators included road network connectivity, access to electricity, population density, and health-worker density. Detailed descriptions of each indicator, including its maximum and minimum value, are provided in the appendix (p 8). Each indicator was regressed individually on the inefficiency score, due to collinearity among indicators. Year fixed effects were included in the regression models to control for non-linear effects of time, and each indicator was standardised to have a mean of 0 and SD of 1. Inefficiency uncertainty was captured by regressing each of the 250 draws of inefficiency scores on these indicators. For indicators that provided standard errors, data uncertainty was incorporated using 250 draws of the indicator variable. Model uncertainty was also incorporated by using draws of the variance–covariance matrix rather than the mean value of each coefficient.

Sensitivity analyses

We conducted several sensitivity analyses to assess the impact of the control variables on the frontier model. First, we fit a frontier that controlled for HIV prevalence as well as SDI and joint PAF. HIV was considered due to its substantial impact on mortality and long-lasting impact on costs, especially in regions with high infection rates. Additionally, we fit a frontier that controlled for SDI only, a frontier that controlled for PAF only, and a frontier that did not control for any covariates. We also tested the sensitivity to fitting a frontier on all countries and years rather than omitting country-years of data with total health expenditure less than $35 per capita. We also tested the sensitivity of our results to the chosen health outcome by estimating a frontier with under-5 mortality (deaths per person-years in the under-5 population) as the health outcome. To assess the robustness of our regression findings, we estimated the regressions separately for the four World Bank income groups: high-income, upper-middle-income, lower-middle-income, and low-income countries. We also explored variation over time in two ways. First, we assessed the efficiency frontier from 1995 to 2009, and 2010 to 2019. Second, we estimated coefficients of a regression model with a COVID-19 interaction, with COVID-19 defined as occurring in the years 2020–22.

This study complies with the GATHER statement, with further information provided in the appendix (pp 5–7). Because we used only secondary, de-identified data, ethical approval was not required for this study.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, the writing of the report, or the decision to submit the manuscript for publication.

Results

In 2022, the population-weighted average distance from the frontier was 3·3 (95% UI 3·0–3·6) health-adjusted life-years (figure 1), meaning that the average person could gain 3·3 years of healthy life if all countries were operating on the frontier. The GBD super-region with the smallest inefficiency in 2022 was southeast Asia, east Asia, and Oceania, which had a population-weighted inefficiency score of 1·0 (0·8–1·2) health-adjusted life-years. The super-region with the largest inefficiency was sub-Saharan Africa, with a population-weighted inefficiency score of 5·8 (5·2–6·5) health-adjusted life-years. Inefficiency scores by super-region are shown in the appendix (p 41).

Inefficiency scores for all countries in 2022 are shown in figure 2 and are listed in the appendix (p 41). The USA has the highest per-capita health-care spending, which corresponds to an above-average inefficiency of 6·2 (95% UI 5·6–6·9) health-adjusted life-years, despite controlling for its high rates of behavioural and metabolic risks. The most inefficient countries in 2022 were Nauru (13·3 [11·9–14·9] health-adjusted life-years), Equatorial Guinea (12·7 [11·5–14·0]), and Eswatini (12·3 [10·9–13·6]). China was the only country to achieve zero inefficiency in 2022 (0·0 [0·0–0·0] health-adjusted life-years).

We found that the average inefficiency for each super-region was decreasing up to 2019 (figure 3). These improvements are measured relative to a single frontier and therefore capture both technical change and efficiency change. The countries that made the largest improvements from 1995 to 2019 were Rwanda, which moved 15·5 (95% UI 13·7–17·7) health-adjusted life-years closer to the frontier, and Burundi, which moved 14·3 (12·7–16·1) health-adjusted life-years closer to the frontier. Inefficiency scores for all countries in 1995 and 2019 are listed in the appendix (p 41). However, some individual countries experienced setbacks. The largest increases in inefficiency were experienced by Paraguay, which moved 3·8 (3·4–4·2) health-adjusted life-years away from the frontier, and Lesotho, which moved 3·7 (2·7–4·9) health-adjusted life-years away from the frontier.

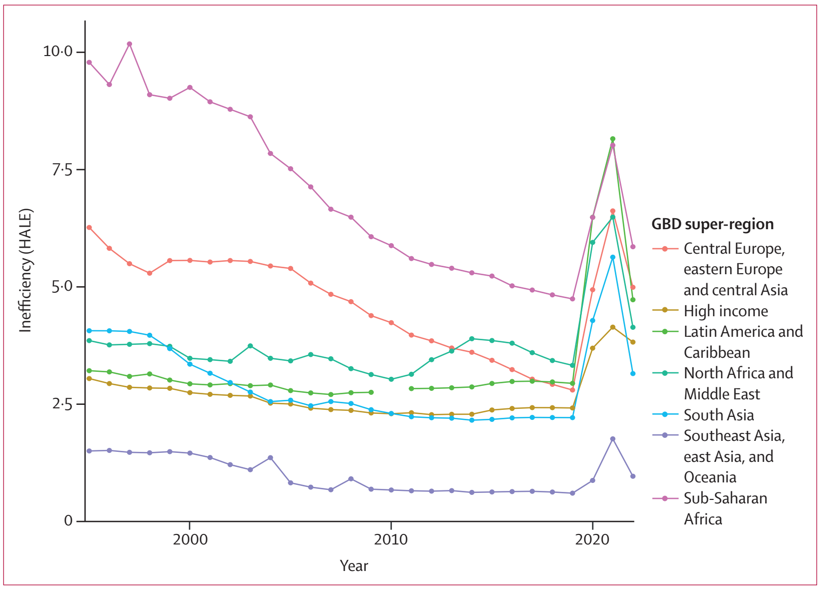

Figure 3: **Inefficiency by GBD super-region, 1995–2022**
For each region, the inefficiency scores are calculated as a population-weighted average of the score of each country in the super-region. GBD regions are made up of countries and territories that are geographically close and epidemiologically similar. GBD regions are grouped into super-regions based on observed cause of death patterns. The 2010 datapoint for the Latin America and the Caribbean super-region is removed due to the transitory shock of the Haiti earthquake. GBD=Global Burden of Diseases, Injuries, and Risk Factors Study. HALE=health-adjusted life expectancy.

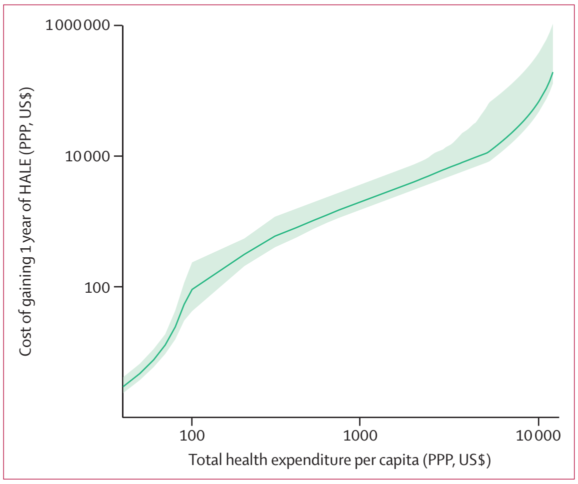

Figure 4: **Cost of increasing HALE by 1 year**
The cost of increasing HALE by 1 year is measured in terms of total health spending per capita. It is calculated as the inverse of the slope of the efficiency frontier. The discontinuities in the cost function arise from the spline knots used to create the frontier. Uncertainly intervals, depicted by the shaded region, are based on the 2·5th and 97·5th percentile of 250 estimates. The solid line represents the mean estimate. HALE= health-adjusted life expectancy. PPP=purchasing power parity.

The COVID-19 pandemic caused inefficiency to spike in 2020 and 2021 globally, meaning countries were generally spending more than or the same amount as before, but with worse outcomes. Across all countries, the average distance from the frontier increased from 2·3 (95% UI 2·1–2·5) health-adjusted life-years in 2019 to 5·0 (4·6–5·4) health-adjusted life-years in 2021. Countries in Latin America and the Caribbean experienced the largest setback during the pandemic, with their inefficiency score rising to 8·1 (7·4–8·8) in 2021.

All GBD super-regions recovered some losses in 2022. Latin America and the Caribbean had the largest improvement from 2021 to 2022, with inefficiency dropping from 8·1 (95% UI 7·4–8·8) to 4·7 (4·2–5·3) health-adjusted life-years. The high-income super-region experienced the least improvement, moving from 4·1 (3·9–4·4) to 3·8 (3·5–4·1) health-adjusted life-years.

The cost of one health-adjusted life-year increased as total health expenditure increased (figure 4). This increase demonstrates the diminishing returns from spending more money at a given level of technology (ie, low-income countries will experience larger gains in health for the same amount of investment). At an annual spending level of around $100 per capita, such as in Ethiopia and Haiti, one additional health-adjusted life-year would cost $92 (95% UI 43–239) per capita. For countries spending approximately $5000 per capita, such as Spain and Slovenia, one additional health-adjusted life-year would cost $11 213 (8031–57 754) per capita (appendix p 48).

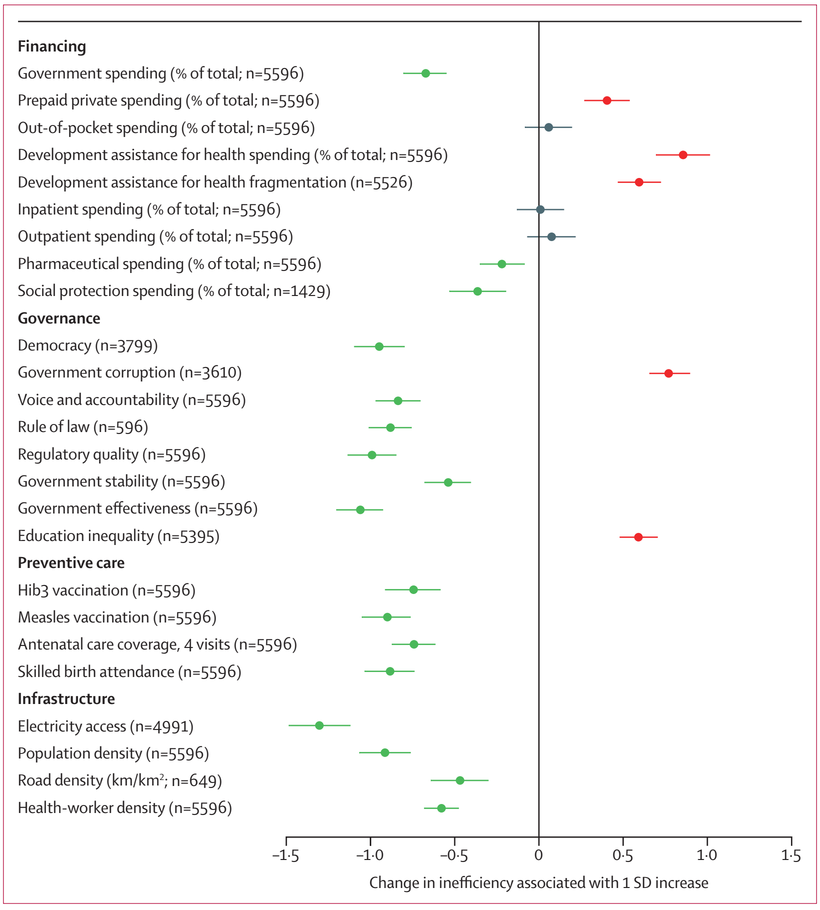

Figure 5: **Factors associated with health spending inefficiency**
The plot shows the change associated with a 1 SD increase in each indicator. Error bars are 95% uncertainty intervals that account for uncertainty in health spending. Statistical significance at the 95% level is shown by green bars (significant decrease in inefficiency) or red bars (significant increase in inefficiency); black bars depict no change in inefficiency. To capture the variable amount of data available for each indicator, the sample size is shown in parentheses for each indicator. Hib3=*Haemophilus influenzae* type B.

Indicators of well-functioning governments were associated with more efficient health spending, while a higher level of corruption was associated with less efficient spending (figure 5). An increase in the democracy index of 1 SD, which indicates a well-functioning and open democracy (6·4 points) was associated with a decrease in inefficiency of 0·9 (95% UI 0·8–1·1) points. Education inequality was also associated with inefficiency, although the strength of the association was smaller. An increase of 1 SD in education inequality (0·2 points) was associated with an increase in inefficiency of 0·6 (0·5–0·7) points.

Having a higher percentage of health expenditure from government spending was associated with more efficient health spending. Conversely, having a higher percentage of health expenditure from development assistance and more fragmented development assistance spending were associated with less efficient health spending. The strength of the associations for financing indicators were smaller than those for governance indicators. An increase of 1 SD in the percentage of total health expenditure from government expenditure (22·3%) was associated with a decrease in inefficiency of 0·7 (95% UI 0·5–0·8) points.

Our results showed that all measures of preventive care uptake (vaccination coverage, antenatal visits, and skilled birth attendance) were associated with more efficient spending. Infrastructure factors that facilitate easier access to care were also associated with decreased inefficiency. Electricity access, road density, population density, and the number of health workers per capita were all associated with more efficient spending. Of these indicators, electricity access had the strongest association with inefficiency, with an increase of 1 SD (30·3%) in access to electricity associated with a decrease in inefficiency of 1·3 (95% UI 1·1–1·5) points.

The sensitivity analyses showed that our primary results were robust to different model specifications and different outcome measurements. A complete set of frontier figures, maps, and regression results are included in the appendix (pp 55–72).

Discussion

This study found that most countries have the potential to make health gains without increasing spending. Based on countries' health outcomes in 2022, eliminating all inefficiency would increase global HALE by 3·3 (95% UI 3·0–3·6) years.

COVID-19 had an almost universal effect of driving up inefficiency in 2020 and 2021. This reflects the disruption caused by the pandemic in health-care system functioning; this effect began to reverse in 2022. Moving forward, it will be crucial to explore the extent to which the long-term impacts of COVID-19 on health-care system inefficiency differ by country, and, importantly, for health-care systems to learn from the impact of the pandemic and invest more wisely in pandemic preparedness.

We found that the returns to additional health spending decrease at higher levels of expenditure. Diminishing returns occur due to the most cost-effective programmes being implemented first, as well as limits to technology as life expectancy increases. For all countries, the most cost-effective way to improve health outcomes is to decrease inefficiency and move closer to the frontier. For low-income countries, increasing health spending will also provide substantial gains in HALE. For high-income countries, achieving gains in HALE by increasing spending becomes prohibitively expensive; therefore, reducing inefficiency or pushing the frontier outward through improvements in technology are the only affordable ways to improve health outcomes.

We found that countries with higher levels of corruption had less efficient health spending, which is in line with previous studies.[5,6] Countries with a higher proportion of health spending coming from the government had more efficient health spending. This aligns with findings from previous studies that higher universal health-care coverage was associated with more efficient health spending, and that non-universal government health-care plans suffer from adverse risk selection, with only the highest-risk populations enrolling.[6,18,19] Broadening the perspective beyond health spending, countries that allocate a higher percentage of government spending to social protection also had more efficient health spending. This is consistent with previous findings, for example, on cash transfers (ie, programmes that provide unrestricted funds with a goal of poverty reduction). Several studies have found health benefits of both conditional cash transfers (programmes in which the payment is conditional on criteria such as children's enrolment in school) and unconditional cash transfers, including increased health-care utilisation and decreased incidence of low birthweight.[20–23]

Countries with health-care budgets that were more reliant on development assistance had less efficient health spending. Among countries that received development assistance, those in which development assistance was more fragmented had less efficient health spending. The relationship between development assistance and inefficiency should be interpreted with caution, given that donors are more likely to donate to countries with worse health outcomes. The finding that highly fragmented aid resulting in wasteful spending is policy-amenable, however, is echoed in a case study based in the Democratic Republic of the Congo, which found that consolidation of development assistance resulted in substantial cost savings.[18] Other studies have also highlighted inadequate harmonisation among donors as a concern.[24,25]

Outside of the health-care system, our results indicate that governments will see health gains by investing in infrastructure that facilitates easier access to health care. This finding is consistent with previous research that has found electricity access to be associated with better health outcomes and linked new roads to an increase in antenatal care.[26,27]

Many other factors related to health-care utilisation, including health behaviours, are important drivers of inefficiency but are difficult to study due to a scarcity of cross-country data availability. Yip and Hafez[18] analysed country-specific case studies and found several sources of inefficient health spending, including underuse of generic medications, inappropriate prescriptions, a suboptimal mix of health workers and infrastructure, and a shortage of long-term care facilities, which led to long hospital stays. A recent report by the Organisation for Economic Co-operation and Development (OECD) found that over-prescription of antibiotics and under-use of generics were also sources of waste in OECD countries.[28] The report highlighted additional inefficiencies including medical errors, use of the emergency department for conditions that could be managed in an outpatient setting, and medically unnecessary procedures such as caesarean sections or imaging.

Although our study does not establish causal relationships, future study of possible mechanisms that could link these factors to health spending inefficiency is warranted. Governance indicators might influence inefficiency through corruption or rent-seeking behaviours that result in increased prices or low-value care. Vaccination programmes save money directly via avoided illness, but high vaccination rates might also indicate a health-care system with good geographical coverage or effective public communication. Road density could lead to better outcomes for time-sensitive causes such as injuries or stroke.

For countries with high levels of inefficiency, gains can come from improvements in either allocative or technical efficiency. To maximise allocative efficiency, funding for the most cost-effective interventions should be prioritised. To maximise technical efficiency, costs for a given intervention should be lowered. Improved data collection on health-care utilisation and spending by health condition would enable future research to examine variation in and drivers of both allocative and technical efficiency.

This study has several limitations. First, the efficiency frontier is based on observed evidence and does not necessarily reflect the theoretical best possible outcome for a given level of spending. In particular, an inefficiency score of zero should not be interpreted as a country having a health-care system with no possible areas of improvement, but rather that it has achieved the best outcome among its peers with similar levels of health spending. Second, the assumption of Gaussian errors in the SFMA model is a limitation. The likelihood-based approach to detect outliers helps to mitigate this issue by removing datapoints that are poorly fit by a Gaussian assumption. Third, health-care systems are multi-dimensional and no single output can capture their full complexity. For example, quality is an important dimension of health-care systems that is not addressed in this work. Depending on the quality measure, an improvement in quality might increase costs or drive down costs, and improvement in a subjective quality measure such as patient experience might not translate to an increase in HALE.[29] Fourth, there are many potential drivers of inefficiency that we were unable to capture due to an absence of data. One key example is the implementation of disease screening programmes, such as blood pressure screenings or mammograms. Our study found that preventive care measures such as vaccination rates were found to be associated with reduced inefficiency, and additional research in this area would be beneficial.

Fifth, the control variables included in the frontier model are not completely exclusive to the health-care system. For example, total fertility rate is a component of SDI and can be affected by access to contraceptives. To address this limitation, we performed a sensitivity analysis in which no covariates are controlled for, to estimate a model in which 100% of the variation in HALE is explained by health expenditure. Sixth, HALE is a slow-moving metric. Although our analysis controls for risk factors and health spending for each year, health outcomes are also dependent on previous exposure to risks and medical treatment. Under-5 mortality is included as a sensitivity analysis to examine inefficiency in the context of a metric that responds quickly to health-care systems changes. The indicators assessed in this work have limitations in terms of data availability and quality. Data availability for each indicator is shown in the appendix (p 13). Data uncertainty of the indicators was incorporated where available, but the level of uncertainty might be underestimated. Last, the associations between inefficiency scores and governance and financing indicators are not causal relationships. These indicators were regressed individually, but many of them are correlated with each other as well as with other unobserved variables. Therefore, the mechanism of effect must be better established to be confident of specific policy recommendations.

This study found that inefficiency in health spending decreased consistently from 1995 to 2019. The COVID-19 pandemic had a large negative impact on health spending efficiency in 2020 and 2021, but ground was regained in 2022. Future work should examine how health spending inefficiency throughout the world continues to evolve in the post-pandemic era.

Countries around the world have made significant progress in converting dollars into health. However, more reductions in inefficiency need to be made in an era of tightening health-care budgets. To maximise the returns on their health-care spending, countries should consider increasing the percentage of their population that is covered by health insurance, work to improve governance, and invest in social protection spending, preventive care, and infrastructure that improves access to health care.

Contributors

AL and MRB prepared the first draft of the manuscript. JLD and SSL were responsible for development of the methods. MRB and AL accessed and verified the data and conducted the data analysis. JLD, CJLM, SIH, and AL all contributed to subsequent of the draft. KVT and JLD provided project management. All authors had full access to all the data in the study and had final responsibility for the decision to submit the manuscript for publication.

Declaration of interests

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Data sharing

Data used for this study were extracted from publicly available sources that are listed in the appendix (p 9). Estimates created in this study will be available to the public indefinitely on the Global Health Data Exchange website at https://ghdx.healthdata.org.

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Editorial note: The Lancet Group takes a neutral position with respect to territorial claims in published maps and institutional affiliations.

References

1 Institute for Health Metrics and Evaluation. Financing global health 2023: the future of health financing in the post-pandemic era. Seattle, WA: IHME. May 14, 2024. https://www.healthdata.org/research-analysis/library/financing-global-health-2023-future-health-financing-post-pandemic-era (accessed May 9, 2025).

2 Mao W, Prizzon A, Ogundeji Y. UK foreign aid cuts and global health. *BMJ* 2023; **382:** 2075.

3 Kates J, Wexler A, Rouw A. The Trump administration's foreign aid freeze and global health. KFF. March 6, 2025. https://www.kff.org/global-health-policy/issue-brief/the-trump-administrations-foreign-aid-freeze-and-global-health-the-biggest-gaps-left-on-the-donor-landscape/ (accessed March 14, 2025).

4 Simar L, Wilson PW. Estimation and inference in two-stage, semi-parametric models of production processes. *J Econom* 2007; **136:** 31–64.

5 Goyeneche L, Bauhoff S. Efficiency of health systems in middle-income countries and determinants of efficiency in Latin America and the Caribbean. *PLoS One* 2024; **19:** e0309772.

6 García-Escribano M, Juarros P, Mogues T. Patterns and drivers of health spending efficiency. SSRN 2022; published online March 31. https://papers.ssrn.com/abstract=4070782 (preprint).

7 Jordi E, Pley C, Jowett M, Abou Jaoude GJ, Haghparast-Bidgoli H. Assessing the efficiency of countries in making progress towards universal health coverage: a data envelopment analysis of 172 countries. *BMJ Glob Health* 2020; **5:** e002992.

8 Ahmed S, Hasan MZ, MacLennan M, et al. Measuring the efficiency of health systems in Asia: a data envelopment analysis. *BMJ Open* 2019; **9:** e022155.

9 Eze P, Idemili CJ, Lawani LO. Evaluating health systems' efficiency towards universal health coverage: a data envelopment analysis. *Inquiry* 2024; **61:** 469580241235759.

10 Pérez-Cárceles MC, Gómez-Gallego JC, Gómez-Gallego M. Environmental factors affecting European and central Asian health-systems' bias-corrected efficiency. *Appl Econ* 2018; **50:** 3432–40.

11 Zheng P, Worku N, Bannick M, et al. Robust nonparametric stochastic frontier analysis. *arXiv* 2024; published online April 4. https://arxiv.org/abs/2404.04301 (preprint).

12 Estellita Lins MP, Moreira da Silva AC, Lovell CAK. Avoiding infeasibility in DEA models with weight restrictions. *Eur J Oper Res* 2007; **181:** 956–66.

13 GBD 2021 Diseases and Injuries Collaborators. Global incidence, prevalence, years lived with disability (YLDs), disability-adjusted life-years (DALYs), and healthy life expectancy (HALE) for 371 diseases and injuries in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet* 2024; **403:** 2133–61.

14 Apeagyei AE, Bisignano C, Elliott H, Hay SI, et al. Tracking development assistance for health, 1990–2030: historical trends, recent cuts, and outlook. *Lancet* 2025; published online July 15. https://doi.org/S0140-6736(25)01240-1.

15 GBD 2023 Demographics Collaborators. Global age-sex-specific all-cause mortality and life expectancy estimates for 204 countries and territories and 660 subnational locations, 1950–2023: a demographic analysis for the Global Burden of Disease Study 2023. *Lancet* (in press).

16 GBD 2019 Healthcare Access and Quality Collaborators. Assessing performance of the Healthcare Access and Quality Index, overall and by select age groups, for 204 countries and territories, 1990–2019: a systematic analysis from the Global Burden of Disease Study 2019. *Lancet Glob Health* 2022; **10:** e1715–43.

17 Schneider MT, Chang AY, Crosby SW, et al. Trends and outcomes in primary health care expenditures in low-income and middle-income countries, 2000–2017. *BMJ Glob Health* 2021; **6:** e005798.

18 Yip W, Hafez R. Improving health system efficiency. Reforms for improving the efficiency of health systems: lessons from 10 country cases. World Health Organization, 2015. https://iris.who.int/handle/10665/185989 (accessed May 15, 2025).

19 Nicholson D, Yates R, Warburton W, Fontana G. Delivering universal health coverage: a guide for policymakers. World Innovation Summit for Health, 2015. https://wish.org.qa/research-report/delivering-universal-health-coverage-a-guide-for-policymakers/ (accessed May 15, 2025).

20 Lagarde M, Haines A, Palmer N. The impact of conditional cash transfers on health outcomes and use of health services in low and middle income countries. *Cochrane Database Syst Rev* 2009; **2009:** CD008137.

21 Amarante V, Manacorda M, Miguel E, Vigorito A. Do cash transfers improve birth outcomes? Evidence from matched vital statistics, program, and social security data. *Am Econ J Econ Policy* 2016; **8:** 1–43.

22 Owusu-Addo E, Cross R. The impact of conditional cash transfers on child health in low- and middle-income countries: a systematic review. *Int J Public Health* 2014; **59:** 609–18.

23 van Daalen KR, Dada S, James R, et al. Impact of conditional and unconditional cash transfers on health outcomes and use of health services in humanitarian settings: a mixed-methods systematic review. *BMJ Glob Health* 2022; **7:** e007902.

24 Shi J, Jin Y, Zheng Z. Addressing global health challenges requires harmonised and innovative approaches to the development assistance for health. *BMJ Glob Health* 2023; **8:** e012314.

25 Moon S, Omole O. Development assistance for health: critiques, proposals and prospects for change. *Health Econ Policy Law* 2017; **12:** 207–21.

26 Ajide KB, Dauda RO, Alimi OY. Electricity access, institutional infrastructure and health outcomes in Africa. *Environ Dev Sustain* 2023; **25:** 198–227.

27 Aggarwal S. The long road to health: healthcare utilization impacts of a road pavement policy in rural India. *J Dev Econ* 2021; **151:** 102667.

28 Organisation for Economic Co-operation and Development. Tackling wasteful spending on health. Jan 10, 2017. https://www.oecd.org/en/publications/tackling-wasteful-spending-on-health_9789264266414-en.html (accessed Nov 20, 2024).

29 Hussey PS, Wertheimer S, Mehrotra A. The association between health care quality and cost: a systematic review. *Ann Intern Med* 2013; **158:** 27–34.